HW12: Scalability

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Case 1: 1D Partitioning (horizontal slabs)

We have 2D domain of size $n \times n$ partitioned into into horizontal slabs of size

$$(n) \times (\frac{n}{p})$$

where p is the number of processors

- Communication complexity : Considering five point stencil each processor needs data from the processor just after and before (bottom and top). Now since there are n points in each row, that many have to be send/received.

  Hence communication complexity is

  $$2(\alpha + n\beta)$$

- Computation complexity : Taking a 5 point stencil, two operations of multiplication and addition for $n \times \frac{n}{p}$ points we get computation complexity

  $$2 \times 5 \times n \times \frac{n}{p} \times \gamma = 10\frac{n^2}{p}\gamma$$

- Efficiency calculation : We can see that for n^2 points the single processor case complexity is $10n^2\gamma$. So efficieny in this case is -

  $$E_p = \frac{\frac{10n^2\gamma}{p}}{2(\alpha + n\beta) + \frac{10n^2\gamma}{p}} = \frac{1}{1 + \frac{p(\alpha+n\beta)}{5n^2\gamma}}$$

  For our proposed machine taking $\alpha = 10^{-6}$, $\beta = 10^{-9}$, $\gamma = 10^{-9}$ and $n = 10^6$ and $p = 10^3$ we get,

  $$E_p = \frac{1}{1 + \frac{1.001}{5000}} \approx 99.98\%$$

Further for weak scaling we have taking constant memory $n^2/p = M$ where M is a constant.

$$\implies E_p = \frac{1}{1 + \frac{\alpha}{5M\gamma} + \frac{\sqrt{p}\beta)}{5\sqrt{M}\gamma}}$$

Now taking $p \to \infty$ we get

$$E_p \approx (\sqrt{p})^{-1}$$

Hence seen that 1-d partitioning doesn't scale weakly

Case 2: 2D Partitioning (square blocks)

We now partition the domain into 2D patches of size:

$$\left(\frac{n}{\sqrt{p}}\right) \times \left(\frac{n}{\sqrt{p}}\right)$$

- Communication complexity : Each processor needs data from all four sides. Now since there are $n/\sqrt{p}$ points on each side that many have to be send/received for each side.

  Hence communication complexity is

  $$4(\alpha + \frac{n}{\sqrt{p}}\beta)$$

- Computation complexity : Taking a 5 point stencil, two operation i.e multiplication and addition for $(\frac{n}{\sqrt{p}} \times \frac{n}{\sqrt{p}})$ points we get computation complexity

  $$2 \times 5 \times (\frac{n}{\sqrt{p}} \times \frac{n}{\sqrt{p}}) \times \gamma = 10\frac{n^2}{p}\gamma$$

- Efficiency calculation : We can see that for n^2 points the single processor case complexity is $10n^2\gamma$. So efficieny in this case is -

  $$E_p = \frac{\frac{10n^2\gamma}{p}}{4(\alpha + \frac{\sqrt{n}}{\sqrt{p}}\beta) + \frac{10n^2\gamma}{p}} = \frac{1}{1 + \frac{2p(\alpha + \beta n/\sqrt{p})}{5n^2\gamma}}$$

  For our proposed machine taking $\alpha = 10^{-6}$, $\beta = 10^{-9}$, $\gamma = 10^{-9}$ and $n = 10^6$ and $p = 10^3$ we get,

  $$E_p = \frac{1}{1 + \frac{3.2 \times 10^{-5}}{5000}} \approx 100.00\%$$

Further for weak scaling we have taking constant memory $n^2/p = M$ where M is a constant.

$$\implies E_p = \frac{1}{1 + \frac{2\alpha}{5M\gamma} + \frac{2\beta}{5\sqrt{M}\gamma}}$$

Now taking $p \to \infty$ we get

$$E_p = \frac{1}{1 + 4 \times 10^{-7}} \approx 100.00\% \implies E_p = constant$$

Hence seen that 2-d partitioning scales weakly

Key Results

	1D Partitioning	2D Partitioning
Communication Time	$2(\alpha + n\beta)$	$4(\alpha + \frac{n}{\sqrt{p}}\beta)$
Weak Scaling	NO	YES
Speedup	$\sim O(\sqrt{p})$	$\sim O(p)$
Efficiency	$\sim O(\sqrt{p})^{-1})$	$\sim O(1)$